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[B.tech CSE (core)
Section- A)

26/2/26
Thursday.

# DBMS ASSIGNMENT-1

~~TEST~~

09/10

Q1) What are the key reasons why a DBMS is preferable to a traditional filesystem? How does it ensure data security and consistency.

Q2) Scenario :- A hospital wants to store patients, Doctor and appointment data in a DBMS.

Q- Identify the entities, attributes and relationships in the system. Draw the ER diagram and explain the key design decision you made.

Q3) Scenario :- An online shopping platform needs to store data about customers orders and products.

Que :- Suggest primary keys, foreign keys and integrity constraints for the system. How do these constraints help maintain the accuracy of data?

Q4) Why is understanding Entity-Relationship models important designing a database? What could happen if a relationship is not correctly defined between entities?

## ANSWERS

Q1) Ans Database management system is a software that allows us to store, manage, retrieve, update data efficiently establishing a relationship between (users, apps) and databases.

- Traditional filesystems are restrictive and arent flexible. because :
- ~~Traditional filesystems~~

1) Data Redundancy :- Data multiplication across databases.

2) Data Inconsistency :- outdated data stored in file systems.

3) Manual Entries :- need to store data manually which is time consuming.

4) No backups :- data can be lost easily providing ~~a~~ se zero security.

1) An DBMS ensures data security and consistency by:-

1) Automatic Backups :- Databases are not lost and are stored in a structured manner.

2) Data consistency :- DBMS provides different types of Database storage systems for different data structures such as (RDBMS), (OODMS) etc.
Ensuring management of raw data, unstructured data, semi-structured data etc.
• DBMS helps easy storage, management, updation and retrieval of data eliminating data inconsistency

3) DBMS ~~ensures~~ allows concurrent access of data.

4) ~~Databases are~~ In DBMS, the data structure is independent of any application.

5) DBMS has integrity constraints ensuring accuracy and consistency.

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Q2) Ans. Entity - Relationship for Hospital

- Entities, attributes and relationships identified :-

1) Entities :- 1) Patient (Patient ID, contactNo, Name, Address).

2) Doctor (DoctorID, Name, contactNo)

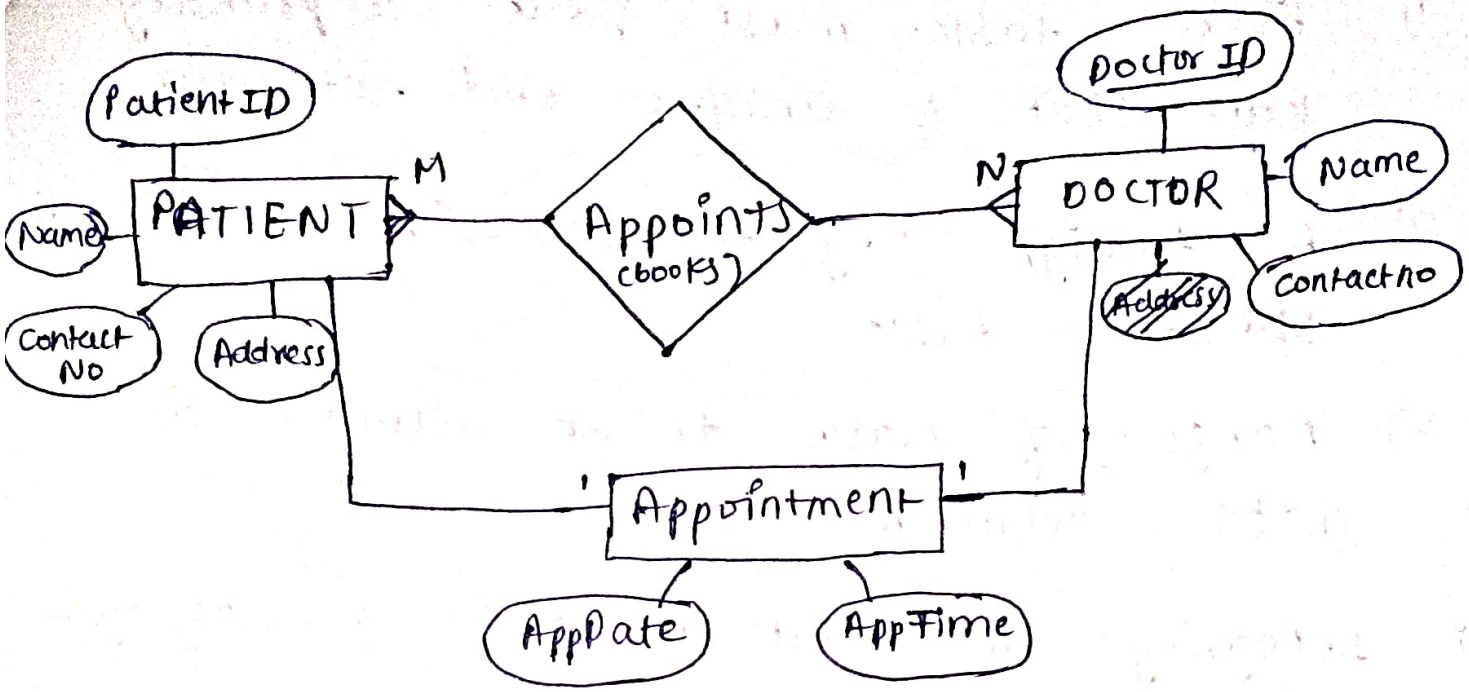
3) Appointment (AppointmentDate, AppTime).

2) Relationships :- 1) Patient books an appointment for their issue.

3) Key design decision :- 1) One patient can have appointments with many doctors.

2) A doctor can have multiple appointments 7 patients.

# ER diagram




## HOSPITAL MANAGEMENT SYSTEM

Q3) online shopping platform to store customers orders and product data.

→ Primary keys, foreign keys and attributes are as follows :-

| | Primary key | Foreign key | Attributes |
|---|---|---|---|
| 1) Customers :- | CustomerID | no key | Name, Address, Email. |
| 2) Products :- | ProductID | no key | Name, price. |
| 3) orders :- | OrderID | CustomerID | TotalAm. Date. |

* Integrity constraints

1) Every table must have a primary key. that is unique and not null

2) A foreign key to establish relationship b/w two tables.

3) Range of data to be allowed in data colouimns

* Integrity constraints maintain accuracy by:-

1) No duplication :- Primary key ensures data uniquenss and prevents accidental duplication of data.

2) Maintains consistency :- upon updation or removal of any data. The remaining data stays unchanged.

3) It ensures data reliability.

Q4) Understanding Entity Relationship is important because:-

:- 1) ER- models provide clear, graphical representation of the system.

2) It helps in identifying logic behind tables, coloumns, primary keys.

3) It serves a blue print for defining the schema.

* If a relationship is not defined correctly then:-

1) data redundancy.

2) loss of data

3) wastage of storage space

4) loss scalability degradation.